

True or false? (unless otherwise specified.)

### Answer grid

|       | 1                        | 2                        | 3                        | 4                        | 5                        | 6                        | 7                        | 8                        | 9                        | 10                       | 11                       | 12                       | 13                       | 14                       | 15                       | B                        |
|-------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| True  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| False | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

1. Let  $x_0 < x_1 < x_2$ . If  $p, q \in \mathcal{P}_2$  satisfy

$$p(x_i) = y_i, \quad q(x_i) = z_i, \quad i = 0, 1, 2,$$

then  $p + q$  interpolates the data  $(x_i, y_i + z_i)$  for  $i = 0, 1, 2$ .

2. For all  $x_0, x_1, y_0, y_1 \in \mathbb{R}$  with  $x_0 \neq x_1$ , there exists a unique  $p \in \mathcal{P}_2$  such that

$$p(x_0) = y_0, \quad p(x_1) = y_1.$$

3. If  $x_0 < \dots < x_n$  and  $\ell_i \in \mathcal{P}_n$  is the  $i$ th Lagrange basis polynomial, then by definition

$$\ell_i(x_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad i, j = 0, \dots, n.$$

4. If  $p, q \in \mathcal{P}_n$  and  $p(x_i) = q(x_i)$  for  $i = 0, \dots, n$ , where  $x_0 < \dots < x_n$ , then  $p = q$ .

5. If  $f \in \mathcal{P}_d$  and  $p_n \in \mathcal{P}_n$  is its interpolant at the  $n + 1$  distinct nodes  $x_0, \dots, x_n$ , with  $n \geq d$ , then  $p_n = f$ .

6. **Interpolation error.** Let  $f \in C^{n+3}[a, b]$ , let  $a \leq x_0 < \dots < x_n \leq b$ , and let  $p_n \in \mathcal{P}_n$  interpolate  $f$  at  $x_0, \dots, x_n$ . Then, for every  $x \in [a, b]$ , there exists  $\xi_x \in (a, b)$  such that

$$f(x) - p_n(x) = \frac{f^{(n+3)}(\xi_x)}{(n+3)!} \prod_{i=0}^n (x - x_i).$$

7. **Interpolation error.** Let  $u(x) = x^4 + 2x^3 + 3x^2 + 4$  and let  $p_3 \in \mathcal{P}_3$  be its interpolant at the points 1, 2, 3, 4. Then

$$u(x) - p_3(x) = (x-1)(x-2)(x-3)(x-4), \quad \forall x \in \mathbb{R}.$$

8. **Convergence.** For every function  $f \in C[-1, 1]$ , if  $p_n$  is the polynomial interpolating  $f$  at the  $n + 1$  equally spaced nodes in  $[-1, 1]$ , including  $-1$  and  $1$ , then

$$\lim_{n \rightarrow \infty} \left( \max_{x \in [-1, 1]} |p_n(x) - f(x)| \right) = 0.$$

9. **Convergence.** Let  $p_n \in \mathcal{P}_n$  be the polynomial interpolating  $f(x) = \sin(2x)$  at the nodes  $x_i = i/n$ , for  $i = 0, \dots, n$ . Then

$$\lim_{n \rightarrow \infty} \left( \max_{x \in [0, 1]} |p_n(x) - f(x)| \right) = +\infty.$$

10. **Convergence estimate.** If  $f \in C[a, b]$  and  $f_n$  is the **piecewise quadratic** interpolant of  $f$  based on a partition of  $[a, b]$  into  $n$  equally large subintervals, then

$$\lim_{n \rightarrow \infty} \left( \max_{a \leq x \leq b} |f(x) - f_n(x)| \right) = 0.$$

11. **Chebyshev polynomials.** Fix  $n \in \mathbb{N}$  and let  $T_n: [-1, 1] \rightarrow \mathbb{R}$  be defined by  $T_n(x) = \cos(n \arccos x)$ . Then  $T_n$  is a polynomial of degree  $n$ .

12. **Chebyshev polynomials.** Assume that  $T_2$  is defined as in the previous item. Then  $T_2(x) = 2x^2 - 3$ . It may be useful to recall that  $\cos(2x) = 2\cos(x)^2 - 1$ .

13. **Chebyshev polynomials.** Let  $x_0 < \dots < x_n$  denote the roots of the Chebyshev polynomial  $T_{n+1}$  on  $[-1, 1]$ . Using these points as interpolation nodes instead of equally spaced nodes can help reduce the maximum interpolation error.

14. **Julia.** Let  $\mathbf{x} = [0, 1, 2]$  and  $\mathbf{y} = [1, 2, 5]$ . The following Julia code uses the Gregory–Newton approach to compute the interpolating polynomial through the three data points  $(0, 1)$ ,  $(1, 2)$ , and  $(2, 5)$ :

```
function p(z)
    return y[1] + (y[2] - y[1]) * z
           + 1/2 * (y[3] - 2y[2] + y[1]) * z * (z + 1)
end
```

15. **Julia.** Let  $\mathbf{x} = [0, 1, 2]$  and  $\mathbf{y} = [1, 2, 5]$ . The following Julia code correctly evaluates the Lagrange interpolating polynomial through these three data points at a point  $z$ :

```
function p(z)
    return (y[1] * (z - x[2]) * (z - x[3]) / ((x[1] - x[2]) * (x[1] - x[3]))
           + y[2] * (z - x[1]) * (z - x[3]) / ((x[2] - x[1]) * (x[2] - x[3]))
           + y[3] * (z - x[1]) * (z - x[2]) / ((x[3] - x[1]) * (x[3] - x[2])))
end
```

16. **Bonus.** Let  $f \in C^\infty[a, b]$ , and let  $f_n$  be the **piecewise linear** interpolant of  $f$  on a uniform mesh with mesh size  $h_n = \frac{b-a}{n}$ . Then

$$\lim_{n \rightarrow \infty} n \times \left( \max_{x \in [a, b]} |f(x) - f_n(x)| \right) = 0.$$

*Justify briefly.*

## Answer key

Questions 1–15: 1 point for a correct answer, 0 otherwise.

1. **True.** For  $i = 0, 1, 2$ ,  $(p + q)(x_i) = p(x_i) + q(x_i) = y_i + z_i$ .
2. **False.** Two points do not determine a unique polynomial in  $\mathcal{P}_2$ .
3. **True.** This is the defining cardinal property of the Lagrange basis.
4. **True.** The difference  $p - q$  has degree at most  $n$  and  $n + 1$  distinct roots.
5. **True.** The difference  $p_n - f$  belongs to  $\mathcal{P}_n$  and has  $n + 1$  distinct roots.
6. **False.** The interpolation remainder uses  $f^{(n+1)}(\xi_x)/(n+1)!$ , not  $f^{(n+3)}(\xi_x)/(n+3)!$ .
7. **True.** Since  $u^{(4)} = 24$ , the interpolation remainder is exactly the product of the four node factors.
8. **False.** Equidistant interpolation need not converge uniformly for every continuous function.
9. **False.** The function  $\sin(2x)$  is analytic, and these interpolants converge uniformly on  $[0, 1]$ , so the limit is 0, not  $+\infty$ .
10. **True.** Piecewise polynomial interpolation on a refining uniform mesh converges uniformly for continuous  $f$ .
11. **True.** For  $n \in \mathbb{N}$ ,  $T_n$  is a polynomial of degree  $n$ .
12. **False.** The identity is  $T_2(x) = 2x^2 - 1$ , not  $2x^2 - 3$ .
13. **True.** Chebyshev nodes generally reduce the maximum interpolation error compared with equally spaced nodes.
14. **False.** The Gregory–Newton factor should be  $z(z - 1)$ , not  $z(z + 1)$ .
15. **True.** The three displayed terms are the Lagrange basis contributions for the three nodes.

### Bonus.

1. On each mesh interval  $[x_i, x_{i+1}]$ , the interpolation remainder gives

$$f(x) - f_n(x) = \frac{f''(\xi_x)}{2}(x - x_i)(x - x_{i+1})$$

for some  $\xi_x \in (x_i, x_{i+1})$ . Since

$$\max_{x \in [x_i, x_{i+1}]} |(x - x_i)(x - x_{i+1})| = \frac{h_n^2}{4},$$

the stated bound follows. Because  $h_n \rightarrow 0$ , the right-hand side tends to zero.